Introduction to neural networks

Generalization

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Roadmap

1. Performance measurement
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Performance measurement

- The central challenge in machine learning is that we must perform well on new, **previously unseen** inputs.

 [1]

 [2]

- The ability to perform well on previously unobserved inputs is called **generalization**.
  - Dataset split
  - Metric
  - Significance

[1] https://www.bbc.com/news/world-us-canada-43604440

[2] https://www.formacar.com/en/tesla-63631-article

Performance measurement

We will introduce a validation step and a test in order to measure generalization.

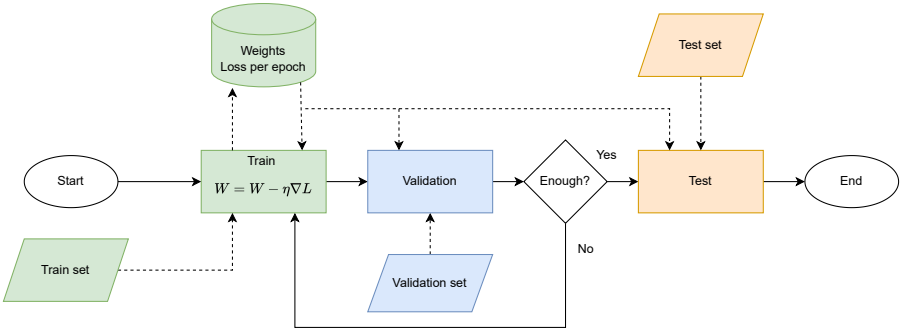


Figure: Train and test workflow

Roadmap

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Dataset split

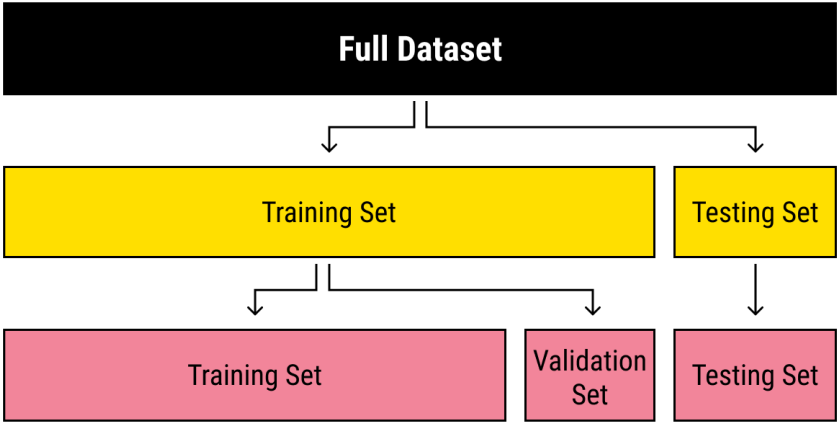


Figure: Dataset split [HJC]

- Lets suppose a dataset, $X_\Omega = \{x_1, \ldots, x_m\}$, with positive examples
- We could use a subset of our set of observations as an estimate of the performance

$$X = x_1, \ldots, x_n$$

$$X' = x_{n+1}, \ldots, x_m$$

, X is the **training set** and X' is usually called the **validation set** [Smola].

- Typically, one uses about 80% of the training data for training and 20% for validation [GoodFellow].

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Performance metrics

Common measurements for categorical outputs

- Accuracy
- Precision
- Recall
- F1 score

Confusion matrix

- Computation:

Ground truth	**Positive Prediction**	**Negative Prediction**
Positive Class	TP	FN
Negative Class	FP	TN

Table: Confusion matrix

- TP + FN sums 100 % of the positive class
- FP + TN sums 100 % of the negative class

Example



Figure: Dataset of animals and no animals. [3]

[3] https://www.labelf.ai/blog/what-is-accuracy-precision-recall-and-f1-score

Prediction concepts

For binary classes[4]

Network input	Prediction	Concept
	'animal'	true positive
	'not animal'	false negative
	'animal'	false positive
	'not animal'	true negative

[4] Consider the class 'animal' as the positive class.

Confusion matrix

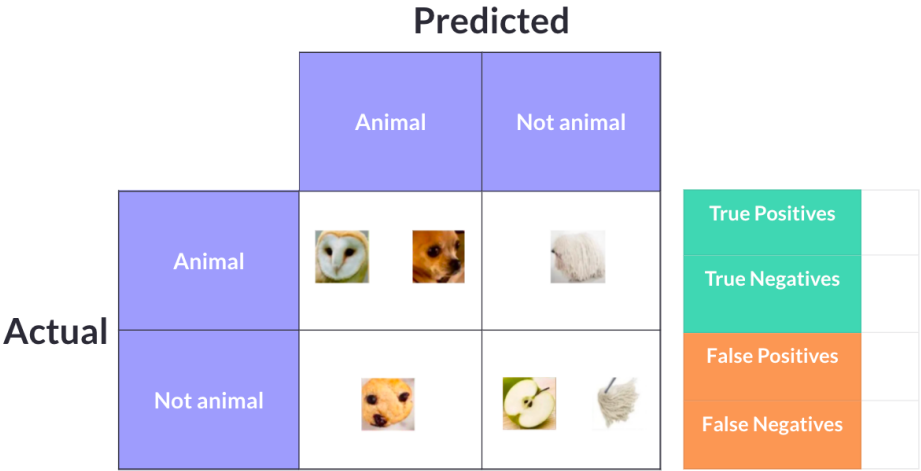


Figure: Confusion matrix example.

Confusion matrix

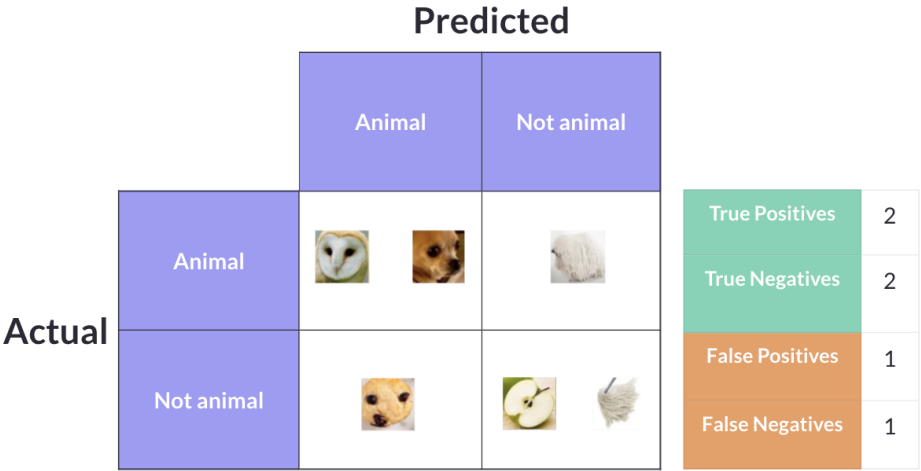


Figure: Confusion matrix example.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

TP represents true positives, TN true negatives, FP false positives, and FN false negatives.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

In this formula, TP represents true positives and FP false positives. This metric measures the proportion of positive identifications that were actually correct.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

TP represents true positives and FN false negatives. Recall measures the model's ability to find all relevant positive instances in the dataset.

The F1-score, which is the harmonic mean of precision and recall, can be expressed as:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

Thus, the F1-score combines both precision and recall into a single metric that accounts for both false positives and false negatives. It is especially useful in situations where the balance between precision and recall is critical.

Significance



Significance



Statistical Significance & Model Evaluation

Definition

A result is statistically significant if the probability of it occurring by chance is very low, typically defined by a threshold α (e.g., 0.05).

Core Concepts:

- **Null Hypothesis (H_0):** No real difference between models.
- **p-value:** $P(\text{Data}|H_0)$. If $p < \alpha$, we reject H_0.

Application in ML (Model Comparison): To compare error rates ϵ_1 and ϵ_2 using a Z-test:

$$Z = \frac{|\hat{\epsilon}_1 - \hat{\epsilon}_2|}{\sqrt{\frac{\hat{\epsilon}(1-\hat{\epsilon})}{n_1} + \frac{\hat{\epsilon}(1-\hat{\epsilon})}{n_2}}} \tag{5}$$

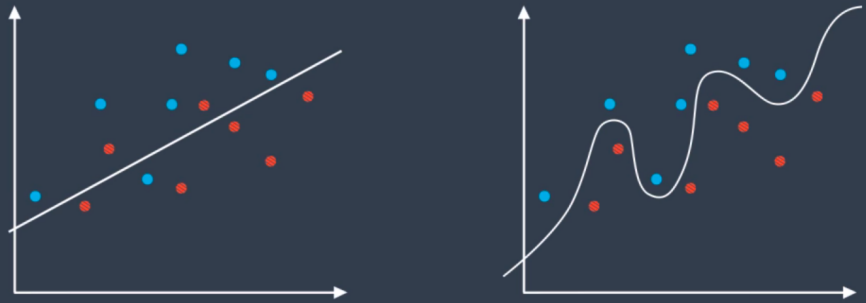
Ensures that performance gains are robust and reproducible.

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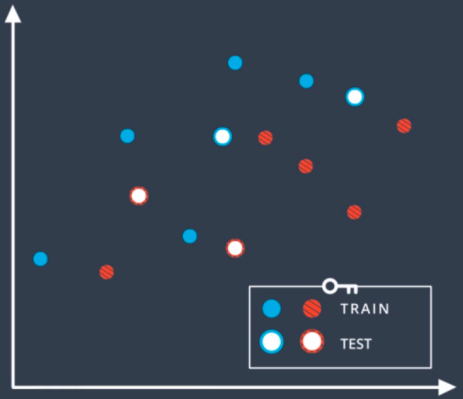
Overfitting problem

Which model is better?



Overfitting problem

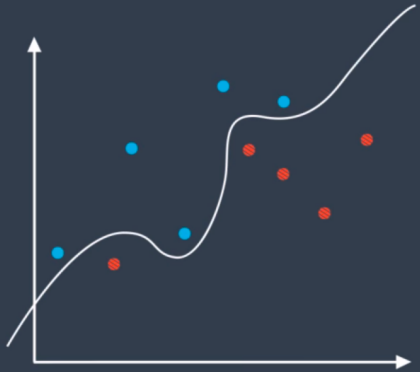
The importance of testing



Overfitting problem

The importance of testing

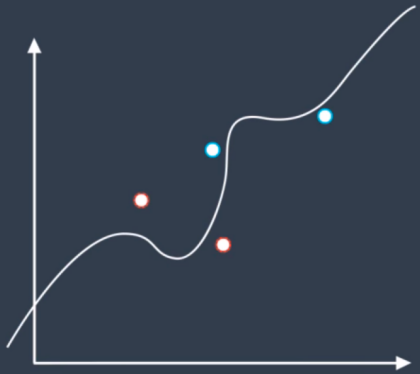




Overfitting problem

The importance of testing





Overfitting and generalization

- **Underfitting** occurs when the model is not able to obtain a sufficiently low error value on the training set.
- **Overfitting** occurs when the gap between the training error and test error is too large.
- A model's **capacity** is its ability to fit a wide variety of functions.

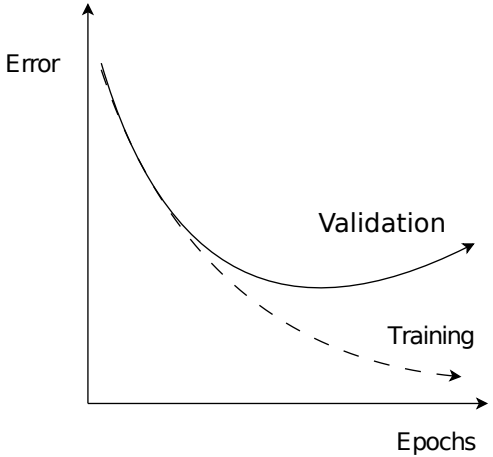


Figure: Loss as epochs advance.

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Regularization

- **Regularization** is any modification we make to a learning algorithm that is intended to reduce its generalization error but not its training error.
  - Early termination
  - L1, L2
  - Dropout

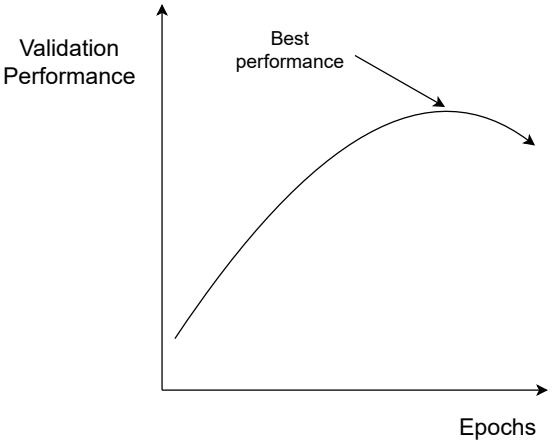


Figure: Performance as epochs advance.

L2 Regularization (Weight Decay)

Adds a penalty equal to the square of the magnitude of coefficients.

The Math:

$$\mathcal{L}(W) = E(W) + \lambda \sum_{j=1}^{n} w_j^2$$

Key Insights:

- Penalizes large weights.
- Leads to **small, non-zero** weights.

L1 Regularization (Lasso)

Adds a penalty equal to the absolute value of the magnitude of coefficients.

The Math:

$$\mathcal{L}(W) = E(W) + \lambda \sum_{j=1}^{n} |w_j|$$

Key Insights:

- Encourages **sparsity** (many weights become exactly zero).
- Useful for **feature selection**.

Dropout

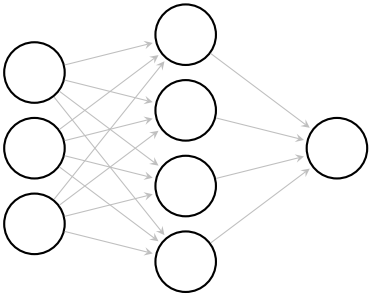
A stochastic technique mainly used in Deep Learning.

- **How it works:** During training, randomly "deactivate" a percentage p of neurons in each forward pass [5].
- **Why it works:**
  - Prevents **co-adaptation** of neurons.
  - Forces the network to learn redundant representations.
  - Acts as an approximate model ensemble (averaging many sub-networks).

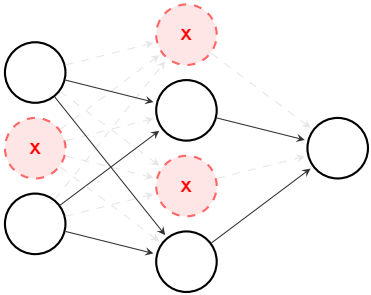
[5] All neurons are used during inference, but weights are scaled by p.

Dropout

Standard Network



Network with Dropout



Data Augmentation

- Artificially increases the size and diversity of the training set by generating new examples from the existing ones.
- Acts as a regularizer: it reduces overfitting and improves generalization, especially with small datasets.
- New samples are created by applying label-preserving transformations:

$$\mathbf{x}' = A(\mathbf{x}; \phi), \qquad y' = y \tag{6}$$

$$\phi \sim p(\phi)$$

Parameters ϕ (e.g., rotation angle, noise level) are sampled randomly at training time. Unlike preprocessing, augmentation is applied **only to the training set**, never to validation/test data.

- **Geometric transforms:** rotation, translation, scaling, horizontal/vertical flips, cropping.
- **Photometric transforms:** brightness, contrast, saturation changes, color jittering.
- **Noise injection:** adding Gaussian noise,

$$\mathbf{x}' = \mathbf{x} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}) \tag{7}$$

- **Erasing/occlusion:** randomly masking regions of the input (random erasing, cutout).
- Transformations must preserve the label: e.g., a vertical flip may be invalid for letters ('A' vs '∀') or digits ('6' vs '9').

Reading and writing

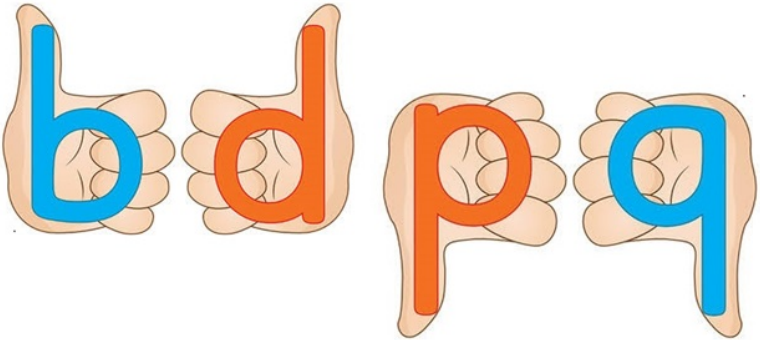


Figure: Rotations and transforms

Model capacity selection

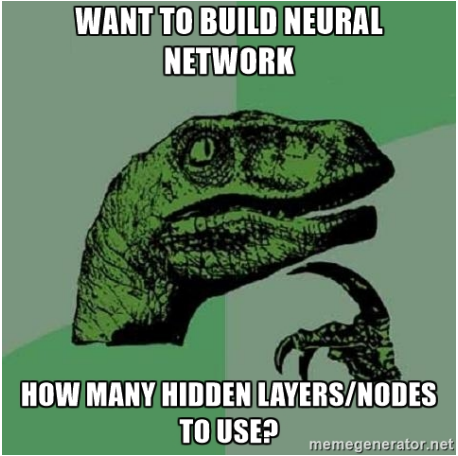
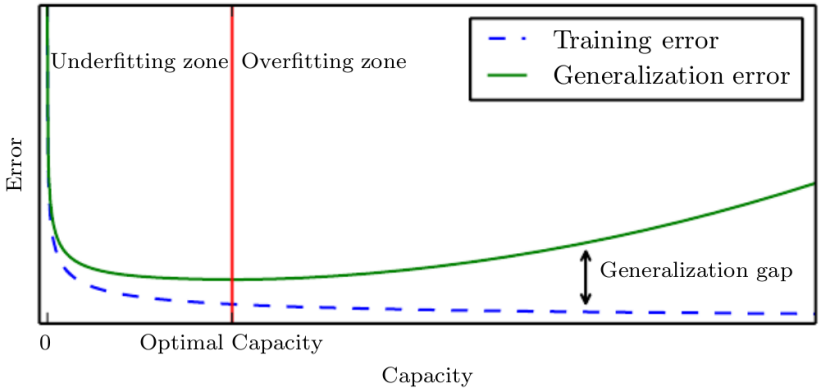


Figure: Selecting depth

Model capacity selection

- Usually select the simplest model
- The optimal model is the one that has the same capacity than the problem.



Aditional information

- Chapter 5 Machine Learning Basics. Ian Goodfellow, et al. Deep Learning. MIT.
- Chapter 2 Training feed-forward neural networks. Buduma. Fundamentals of deep learning.

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